	B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19 **Computer Science & Engineering**	
INTERNALS-1		
Course Code: 23CS3PCDST	**Course Title: Data Structures**	
Semester: 3	**Maximum Marks: 40**	**Date: 29/10/2025**
Faculty Handling the Course:	**SHS,LN,Dr.SKS,ANS, Dr.NM,Dr.MVS,Dr.HSG,SA, SPN,MSN,PP,BSS,MS,PCV**	
Instructions: Internal choice provided in Part C.		

PART-A

Total 5 Marks (No Choice)

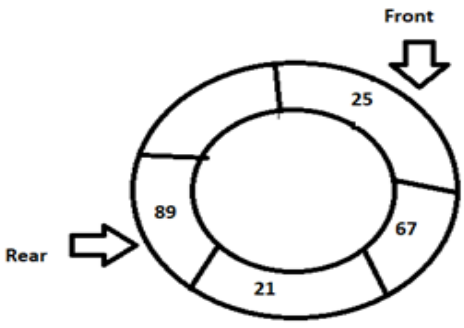
Q.No.	Question	Marks	CO-PO
1.	Define data structure. Briefly explain how data structures can be classified	5	1-1

PART-B

Total 15 Marks (No Choice)

Q.No.	Question	Marks	CO-PO
2 a)	Evaluate the following postfix expression. Show the stack tracing contents at every stage: 5 3 2 * + 4 - 5 +	**5**	**2-2**

2 b)	Analyze the given recursive routine. Write the tracing and expected output for the call display (4).
	
	`int display (int n)`
	`{`
	`if (n <= 0)`
	`return 0;`
	`else`
	`{`
	`printf (“%d”, n);`
	`return display (n-1) + display (n-2);`
	`}`
	`}`
		5	**2-2**

2 c)	Assume the initial status of the queue as shown in the diagram. Perform the following operations and show the status of the queue after each operation: 1.Enqueue(90) 2.Enqueue(20) 3.Dequeue() 4.Enqueue(15) 5.Enqueue(40)  	**5**	**2-2**

PART- C

Total 20 Marks

Q. No.	Question	Marks	CO-PO
3 a)	Outline an algorithm to convert an infix expression to a postfix expression using a stack. Convert the following infix expression to postfix expression ((A + B) * C - (D - E) ^ (F + G)).	10	2-2
	OR		
3 b)	The software firm needs to reallocate three critical resources from **Server A** to **Server C** using **Server B** as an intermediate node. Each resource has dependency constraints, and only one can be moved at a time. A large resource can't be placed above a smaller resource. Formulate the sequence of transfers required to complete the reallocation successfully. Write the C program for the above and trace the same.	10	2-2
4a)	An airport parking bay can accommodate a maximum of 4 planes at any given time. As planes arrive, each requests landing clearance along with an emergency level. The airport control system organizes the planes using a certain data structure, in which insertion is based on emergency level, but removal follows the first-in, first-out principle among planes that are parked in the bay. The airport control system ensures that planes with higher emergency levels are given priority in landing and departure operations. Analyze the scenario and Identify which data structure would be most suitable. Write a C program with the following functions to simulate this system: InsertPlane()-To add a plane to the parking bay based on its emergency priority. RemovePlane()-To remove a plane from the parking bay when it departs (removal is done from the front). DisplayPlanes()-To display the names and priorities of all planes currently in the parking bay.	10	3-3
	OR		
4b)	An amusement park has a merry-go-round ride with 5 seats, numbered 0 to 4. Passengers arrive with a token number and take the next available seat. When a passenger leaves, the next arriving person is seated immediately in that position, maintaining the order. Initially, all seats are empty. Identify the data structure that is suitable for this scenario. Write a C program with the following functions to simulate this system: add_passenger(), remove_passenger() and display(). The display function should display the token numbers of the passengers currently on the ride.	10	3-3

*****ALL THE BEST*****